

$$\begin{array}{ccccccc}
 A & \xrightarrow{\quad} & F(A) & \xrightarrow{\quad \nu_A \quad} & G(A) & \xleftarrow{\quad} & A \\
 \downarrow f & & \downarrow F(f) & & \downarrow G(f) & & \downarrow f \\
 A' & \xrightarrow{\quad} & F(A') & \xrightarrow{\quad \nu_{A'} \quad} & G(A') & \xleftarrow{\quad} & A'
 \end{array}$$